

ASSIGNMENT 4:

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CLASS : SY BTECH IT

INHERITANCE AND POLYMORPHISM

/* 4.B) Method Overriding

Develop an application for a social media

platform showcasing method overriding. Implement a base class "Post" with methods for posting and commenting, and derived classes representing different post types (e.g., text post, image post) overriding these methods with specific functionality.

*/

```
import java.util.*;
```

```
// Base class
```

```
class Post {
```

```
    int likes = 1000;
```

```
    // Constructor
```

```
    Post() {
```

```
        System.out.println("Post constructor called");
```

```
    }
```

```
    // Method for posting
```

```
    // overridden by child classes
```

```
    void posting() {
```

```
        System.out.println("Post is added.");
```

```
    }
```

```
    void commenting() {
```

```
        System.out.println("This is a comment section");
```

```
    }
```

```
}
```

```
// Child class of Post
```

```
class text_post extends Post {
```

```
// Constructor of Text Post
text_post() {

    // Calls the constructor of the parent class
    super();

    System.out.println("Text Post constructor");
}

// Method overriding
// This method provides specific functionality for a text post
void posting() {
    System.out.println("This is a Text post");

    // Calls the posting() method of the parent class
    super.posting();
}

// Additional method specific to Text Post
void info() {
    System.out.println("The post will be in text or String format");
}
}

// Another child class of Post
// Demonstrates inheritance and method overriding
class image_post extends Post {

    // Overriding the posting() method
    // Provides functionality specific to an image post
    void posting() {
        System.out.println("This is a Image post");
    }

    // Additional method specific to Image Post
    void info() {
        System.out.println("The post will be in Image format or a jpg file");
    }
}
```

```
}
```

```
// Main class
```

```
public class Main {
```

```
    public static void main(String[] args) {
```

```
        // Creating Scanner object
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.println("Text Post Class: ");
```

```
        // Creating an object of Text Post
```

```
        text_post tp = new text_post();
```

```
        // Calls the overridden posting() method of Text Post
```

```
        tp.posting();
```

```
        // Calls Text Post's own info() method
```

```
        tp.info();
```

```
        // Calls the inherited commenting() method from Post
```

```
        tp.commenting();
```

```
        System.out.println("\nImage Post Class");
```

```
        // Creating an object of Image Post
```

```
        image_post ip = new image_post();
```

```
        // Calls the overridden posting() method of Image Post
```

```
        ip.posting();
```

```
        // Calls Image Post's own info() method
```

```
        ip.info();
```

```
        // Calls the inherited commenting() method
```

```
        tp.commenting();
```

```
System.out.println("\nAccess Sub class using Base class:");
```

```
// Parent class reference
Post p;
p = new text_post();
p.posting();
System.out.println(p.likes);
p = new image_post();
p.posting();
}
}
```

OUTPUT:

```
C:\Users\DIYA\Desktop>javac Main.java

C:\Users\DIYA\Desktop>java Main
Text Post Class:
Post constructor called
Text Post constructor
This is a Text post
Post is added.
The post will be in text or String format
This is a comment section

Image Post Class
Post constructor called
This is a Image post
The post will be in Image format or a jpg file
This is a comment section

Access Sub class using Base class:
Post constructor called
Text Post constructor
This is a Text post
Post is added.
1000
Post constructor called
This is a Image post

C:\Users\DIYA\Desktop>_
```